

Python Data Types and Structures

Python provides fundamental data types for storing different kinds of information. Numeric types include integers, floating-point numbers, and complex numbers. Strings represent sequences of characters with rich manipulation capabilities. Boolean values (True and False) support logical operations. Collections in Python are powerful and flexible: lists are ordered, mutable sequences that can contain elements of different types; tuples are immutable sequences providing data integrity and performance benefits; sets are unordered collections of unique elements useful for membership testing and eliminating duplicates; dictionaries are key-value pairs enabling efficient data lookup and organization. Python 3.7 ensures dictionaries maintain insertion order. Each data type has associated methods providing operations suited to its purpose. Understanding these structures is fundamental to effective Python programming, as they form the building blocks for more complex applications. Python's flexibility with these types enables rapid prototyping and development without verbose type declarations.